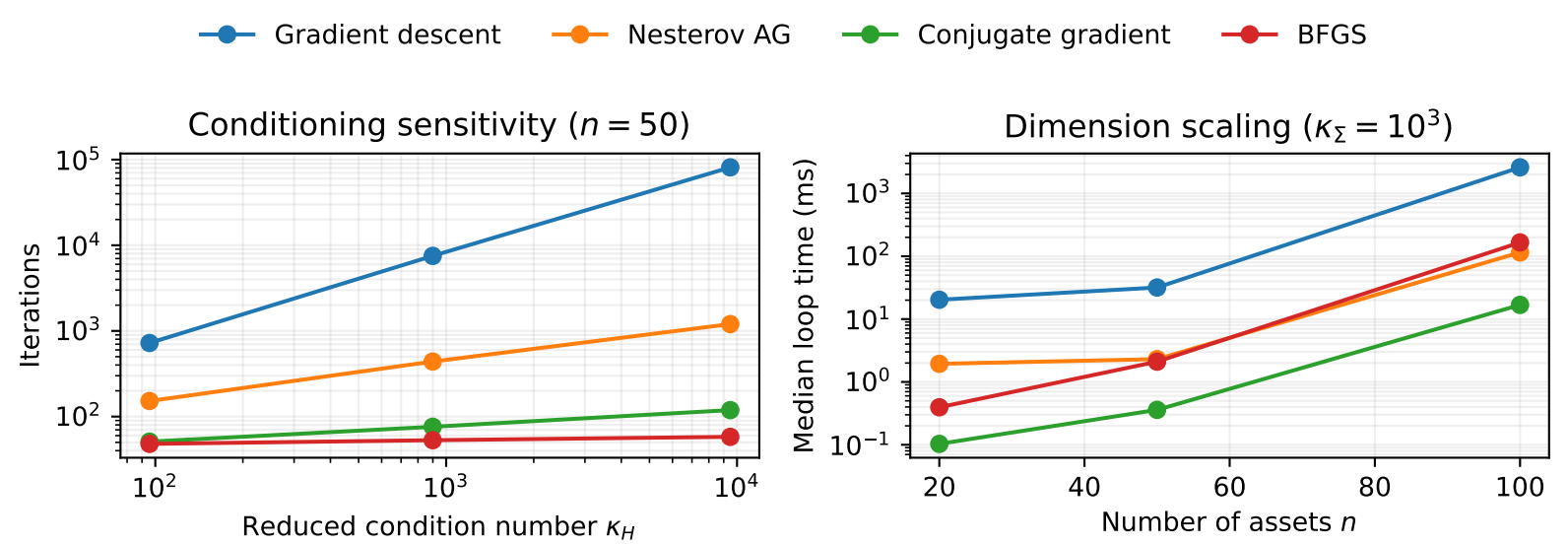
Gradient descent
Nesterov AG
Conjugate gradient
BFGS
Conditioning sensitivity ($n = 50$)
Iterations
10^5
10^4
10^3
10^2
10^2
10^3
10^4
Reduced condition number κ_H
Dimension scaling ($\kappa_\Sigma = 10^3$)
Median loop time (ms)
10^3
10^2
10^1
10^0
10^{-1}
20
40
60
80
100
Number of assets n